

Endpoint permanent gates and cycle-forest families for the Lih–Wang chord inequality

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Abstract

Let J_n be the $n \times n$ matrix whose entries are $1/n$. The Lih–Wang inequality asks whether

$$\text{per}(tJ_n + (1-t)A) \leq t \text{per}(J_n) + (1-t) \text{per}(A), \quad \frac{1}{2} \leq t \leq 1,$$

holds for every doubly stochastic matrix A . We prove a sufficient endpoint criterion which applies to the larger class of nonnegative row-stochastic matrices: if

$$\text{per}(A) \geq 2^{2-n},$$

then the inequality holds throughout the indicated half interval. The proof combines a universal one-collision rook envelope, an analytic estimate for $n \geq 32$, and an exact rational Bernstein certificate for $2 \leq n \leq 31$.

The criterion yields a direct-sum theorem requiring only permanent lower bounds on the blocks. We then study matrices obtained by placing weighted directed cycles on a cycle–element incidence forest. Entropy gives the required endpoint bound whenever there are at least two components (counting fixed points). For a connected incidence tree we prove the inequality for all $n \geq 64$ by a one-hot negative-dependence comparison and an exponential lower-tail estimate. Together with exact results for single cycles and shared-center cycle families, this gives a unified theorem for a natural collection of cycle products and cycle forests. The result does not settle the full Lih–Wang conjecture, and no optimality claim is made for the endpoint threshold.

1 Introduction

For an $n \times n$ matrix $A = (a_{ij})$, write

$$\text{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i,\sigma(i)}.$$

Let Ω_n denote the set of doubly stochastic matrices and let $J_n = \mathbf{1}\mathbf{1}^\top/n$. Lih and Wang asked whether, for every $A \in \Omega_n$,

$$\text{per}(tJ_n + (1-t)A) \leq t \frac{n!}{n^n} + (1-t) \text{per}(A), \quad \frac{1}{2} \leq t \leq 1. \quad (1.1)$$

The conjecture was proved by Lih and Wang in order three. Subsequent work contains several low-order and structurally restricted results. In particular, the complete order-four doubly

stochastic case and a conditional order-six range appear in the published work of Divya K. U. and Somasundaram. Star-matrix results of Arulraj and Somasundaram and of Divya and Somasundaram cover certain direct sums of 2×2 blocks, under hypotheses on the comparison matrix and the block parameters.

The present paper develops a different sufficient mechanism. We fix the comparison matrix J_n , retain only the half interval in (1.1), allow A to be merely row-stochastic, and impose one scalar condition on its permanent. The main endpoint gate is the following.

Theorem 1.1 (endpoint permanent gate). *Let $n \geq 2$, and let A be a nonnegative row-stochastic $n \times n$ matrix. If*

$$p := \text{per}(A) \geq 2^{2-n}, \quad (1.2)$$

then, for every $0 \leq u \leq 1/2$,

$$\text{per}(uA + (1-u)J_n) \leq (1-u)\frac{n!}{n^n} + u \text{per}(A). \quad (1.3)$$

Here $u = 1 - t$, so (1.3) is (1.1) on the original half interval. The theorem does not assert that (1.2) is necessary or optimal.

The block consequence is deliberately weaker than a star-matrix theorem: the comparison endpoint is fixed to J_n , and only the half interval is considered.

Corollary 1.2 (endpoint-controlled direct sums). *Let*

$$A = I_f \oplus A_1 \oplus \cdots \oplus A_b,$$

where A_j is a nonnegative row-stochastic matrix of order n_j , $\text{per}(A_j) \geq 2^{1-n_j}$, and $b + f \geq 2$. Then A satisfies (1.3) for all $0 \leq u \leq 1/2$.

We next turn to cycle-supported matrices. Let C_i be directed cycle permutation matrices, of lengths at least two, and let

$$A = I + \sum_i a_i(C_i - I), \quad a_i \geq 0. \quad (1.4)$$

At every element we require the sum of weights of incident cycles to be at most one; this ensures nonnegativity and row stochasticity. Form the bipartite incidence graph whose vertices are the cycles and the underlying elements and whose edges record membership. Our broad structural result is as follows.

Theorem 1.3 (cycle-forest range). *Assume that the cycle–element incidence graph in (1.4) is a forest. Let c be the number of its connected components containing a cycle, and let f be the number of additional fixed points. Then (1.3) holds in each of the following cases:*

1. $c + f \geq 2$, for every n ;
2. $n \geq 64$, with no restriction on $c + f$;
3. in the remaining connected range $c = 1, f = 0, n < 64$, when there is a single cycle, or when all cycles share one center and their remaining supports are pairwise disjoint.

The theorem does not include incidence graphs containing cycles, and it leaves open connected multi-center forests of order below 64.

Two component theorems used in case 3 are of independent structural interest. One allows pairwise disjoint cycle blocks with independent parameters; the other allows an arbitrary number of cycles sharing one center.

The contribution boundary is important. Orders three and four have prior general doubly stochastic coverage, and parts of the 2×2 -block domain overlap conditional star-matrix results. We make no global-first or strict-extension claim. Our precise literature statement is only that, among the original sources examined, we did not find the combination of the row-stochastic endpoint gate and the full cycle-forest ranges stated above. Several older papers are available to us only through publisher abstracts and reliable later restatements; Section 7 records this limitation.

2 Rook polynomials and the J_n -mixture

Let the rows of a nonnegative row-stochastic matrix A choose columns independently, with row r choosing v with probability a_{rv} . Write N_v for the occupancy of column v , and define the rook polynomial

$$R_A(z) = \mathbb{E} \prod_{v=1}^n (1 + N_v z) = \sum_{k=0}^n h_k z^k. \quad (2.1)$$

The coefficient h_k is the sum of the permanents of all $k \times k$ submatrices obtained by choosing k rows and k columns. Expanding the permanent according to which rows use A and which use J_n gives

$$\text{per}(uA + (1-u)J_n) = \sum_{k=0}^n u^k (1-u)^{n-k} \frac{(n-k)!}{n^{n-k}} h_k. \quad (2.2)$$

It is useful to introduce the positive linear functional

$$\Phi(P) = 2^{-n} \sum_{k=0}^n [z^k] P(z) \frac{(n-k)!}{n^{n-k}}. \quad (2.3)$$

Thus $\Phi(R_A) = \text{per}((A + J_n)/2)$. Put

$$B = 1 + z, \quad F_1 = (1 + 2z)B^{n-2}, \quad c_n = \frac{n!}{n^n}, \quad z_n = 2^{1-n}. \quad (2.4)$$

Lemma 2.1 (universal one-collision envelope). *If $p = \text{per}(A)$, then*

$$R_A(z) \preceq pB^n + (1-p)F_1, \quad (2.5)$$

where $P \preceq Q$ means coefficientwise domination.

Proof. The event that all column occupancies are at most one has probability p , and on that event $\prod_v (1 + N_v z) = B^n$. On every remaining outcome, the occupancy vector has total sum n , at most $n-1$ occupied columns, and at least one collision. Among such integer occupancy vectors,

every elementary symmetric sum is maximized by $(2, 1, \dots, 1, 0)$. Its generating polynomial is F_1 . Averaging gives (2.5). □

The next lemma is the exact bridge from a midpoint estimate to the full half interval.

Lemma 2.2 (midpoint equivalence for large order). *Let $n \geq 32$, let A be nonnegative and row stochastic, and assume $p = \text{per}(A) \geq z_n$. Then (1.3) holds for every $0 \leq u \leq 1/2$ if and only if it holds at $u = 1/2$.*

Proof. Define

$$\beta_k = \frac{(n-k)!h_k}{n^{n-k}\binom{n}{k}}, \quad \gamma_j = 2^{-j} \sum_{k=0}^j \binom{j}{k} \beta_k. \quad (2.6)$$

For every choice of k rows, the probability that their independently selected columns are distinct is at most one. Hence $h_k \leq \binom{n}{k}$, and therefore $\gamma_j(A) \leq \gamma_j(I_n)$. If $\Phi_I = \text{per}((I_n + J_n)/2)$, direct summation gives

$$\gamma_{n-1}(I_n) = z_n, \quad \gamma_{n-2}(I_n) = g_n := \frac{4}{n-1}(\Phi_I - z_n). \quad (2.7)$$

The sequence $j \mapsto \gamma_j(I_n)$ is convex on $0 \leq j \leq n-2$. Indeed, if T has the Gamma($n+1, 1$) law, then

$$\gamma_j(I_n) = c_n \mathbb{E} \left[\left(\frac{1+n/T}{2} \right)^j \right],$$

and its second discrete difference is the expectation of a nonnegative quantity. Moreover

$$2^n \Phi_I \leq 2 + \sqrt{\pi n/2}$$

To see the displayed bound on Φ_I , write its d -th summand as $\prod_{h=0}^{d-1} (1 - h/n)$. For $d \geq 1$,

$$\prod_{h=0}^{d-1} (1 - h/n) \leq \exp\{-d(d-1)/(2n)\} \leq \exp\{-(d-1)^2/(2n)\}.$$

The decreasing Gaussian sum is at most its first term plus its integral, which gives $2^n \Phi_I \leq 2 + \sqrt{\pi n/2}$. Substitution in (2.7) reduces the second assertion below to $8\pi n^3 \leq (n-1)^2(n-2)^2$. At $n = 32$ this follows from $\pi < 22/7$, and $n(1-1/n)^2(1-2/n)^2$ is increasing for $n \geq 3$. Hence, for $n \geq 32$,

$$g_n \leq \frac{n-2}{2n} z_n. \quad (2.8)$$

Convexity between $j = 0$ and $j = n-2$ therefore proves all Bernstein layers $0 \leq j \leq n-2$ of the chord inequality.

It remains to control $j = n-1$. From the envelope (2.5),

$$\gamma_{n-1}(A) \leq p z_n + (1-p) g_n.$$

The difference between the target layer and this upper bound is

$$D(p) = \frac{n+1}{2n}c_n - g_n + p \left(\frac{n-1}{2n} - z_n + g_n \right).$$

Its coefficient of p is positive, and (2.8) gives

$$D(p) \geq D(z_n) \geq \frac{n+1}{2n}c_n + \frac{z_n}{2n}(1 - (n+2)z_n) > 0.$$

Thus the first n Bernstein layers are automatic. The top layer is exactly

$$\gamma_n(A) = \text{per} \left(\frac{A + J_n}{2} \right),$$

so its inequality is precisely the midpoint inequality. Positivity of the degree- n Bernstein basis on $0 \leq 2u \leq 1$ proves sufficiency. Necessity is immediate by setting $u = 1/2$. $\square$

Lemma 2.3 (the finite orders). *For $2 \leq n \leq 31$, the envelope (2.5) implies (1.3) whenever $p \geq 2^{2-n}$.*

Proof. For a polynomial P , set

$$\Gamma_j(P) = 2^{-j} \sum_{k=0}^j \frac{\binom{j}{k}}{\binom{n}{k}} \frac{(n-k)!}{n^{n-k}} [z^k]P. \quad (2.9)$$

After the substitution $x = 2u$, the degree- n Bernstein coefficients reduce to

$$s_{n,j} = \frac{j}{2n} - \Gamma_j(B^n) + \Gamma_j(F_1) > 0, \quad (2.10)$$

$$d_{n,j} = \left(1 - \frac{j}{2n}\right) c_n + \frac{j}{2n} 2^{2-n} - \Gamma_j(2^{2-n}B^n + (1 - 2^{2-n})F_1) > 0. \quad (2.11)$$

The zeroth layer is an equality. All 495 remaining pairs (n, j) , $2 \leq n \leq 31$, $1 \leq j \leq n$, were evaluated in exact rational arithmetic by the frozen endpoint checker. The certificate digest is 9b857ec7f67ca6319d3cab31c2b3f8fd88690766b51898ffbb577c155a9107e.

The smallest margin occurs at $(n, j) = (31, 1)$ and is

$$\frac{550476113801152858212168551495168251918862401}{18327886165296381817380980351835033630345588173537542144} > 0.$$

This certifies a finite symbolic complement, not a sampling of matrices or parameters. $\square$

3 Proof of the endpoint gate

We prove Theorem 1.1. The finite range $2 \leq n \leq 31$ follows from Lemma 2.3, since $p \geq 2^{2-n}$.

Now assume $n \geq 32$, and set

$$\Phi_I = \text{per} \left(\frac{I_n + J_n}{2} \right), \quad C_n = 2 + \sqrt{\frac{\pi n}{2}}. \quad (3.1)$$

Direct evaluation gives

$$0 \leq \Phi(F_1) = z_n - \frac{\Phi_I - z_n}{n-1} \leq z_n, \quad \Phi_I \leq 2^{-n} C_n. \quad (3.2)$$

The lower inequality $\Phi_I \geq z_n$ also follows directly from

$$2^n \Phi_I = \sum_{d=0}^n \prod_{h=0}^{d-1} \left(1 - \frac{h}{n}\right) \geq 2. \quad (3.3)$$

At $u = 1/2$, subtracting the envelope (2.5) from the chord yields the margin

$$G(p) = \frac{c_n}{2} - \Phi(F_1) + p \left(\frac{1}{2} - \Phi_I + \Phi(F_1) \right). \quad (3.4)$$

For $n \geq 32$, $\Phi_I < 1/1024$, so G is increasing in p . Hence

$$G(p) \geq G(2z_n) \geq \frac{c_n}{2} - 2z_n \Phi_I \geq \frac{c_n}{2} - 4 \cdot 4^{-n} C_n. \quad (3.5)$$

The last quantity is positive. Indeed, the integral bound $\log(n!) \geq n \log n - n + 1$ gives $c_n > 3^{-n}$, since $e < 3$. It is therefore enough that $(4/3)^n \geq 8C_n$. At $n = 32$, $C_{32} < 10$ and $(4/3)^{32} > 80$; moreover

$$\frac{C_{n+1}}{C_n} \leq \sqrt{\frac{n+1}{n}} < \frac{4}{3},$$

so the inequality persists for all larger n . Lemma 2.2 now recovers the entire interval $0 \leq u \leq 1/2$. This proves Theorem 1.1. $\square$

Proof of Corollary 1.2. Let $n = f + \sum_j n_j$. Multiplicativity of the permanent under direct sums gives

$$\text{per}(A) = \prod_{j=1}^b \text{per}(A_j) \geq 2^{b+f-n} \geq 2^{2-n}.$$

Apply Theorem 1.1. Notice that no local chord inequality is assumed for any block. $\square$

4 Independent cycle blocks and shared-center families

We record the two low-component inputs used later.

Theorem 4.1 (independent cycle product). *Let $C_1, \dots, C_r$ be pairwise disjoint cycle permutation blocks of lengths $\ell_i \geq 2$, let $f \geq 0$, and put*

$$n = f + \sum_i \ell_i$$

and

$$A = I_f \oplus \bigoplus_{i=1}^r ((1 - s_i)I_{\ell_i} + s_i C_i), \quad 0 \leq s_i \leq 1.$$

Then A satisfies (1.3) for all $0 \leq u \leq 1/2$.

Proof. Set $y_i = (1 - 2s_i)^2$, and define

$$c_\ell(v) = 2^{1-\ell} \sum_{q=0}^{\lfloor \ell/2 \rfloor} \binom{\ell}{2q} v^q.$$

The rook polynomial is

$$R(z; \mathbf{y}) = (1+z)^n \prod_{i=1}^r c_{\ell_i} \left(\frac{1+2z+y_i z^2}{(1+z)^2} \right). \quad (4.1)$$

For the coefficient polynomial of y^q in a single block, write

$$F_{\ell,q}(z) = [y^q](1+z)^\ell c_\ell \left(\frac{1+2z+y z^2}{(1+z)^2} \right), \quad a_{\ell,q} = [z^\ell] F_{\ell,q}.$$

Define $H_0 = 1$ and $H_m = F_{m,0}/2^{1-m}$ for $m \geq 1$. If

$$X = 1 + z + \sqrt{1+2z}, \quad Y = 1 + z - \sqrt{1+2z},$$

then $XY = z^2$ and $H_m = (X^m + Y^m)/2$. Consequently

$$H_a H_b = \frac{1}{2} \left(H_{a+b} + z^{2b} H_{a-b} \right) \preceq H_{a+b} \quad (a \geq b), \quad (4.2)$$

and

$$[z^{m-d}] H_m = 2^d \frac{m}{m+d} \binom{m+d}{2d}. \quad (4.3)$$

For $q \geq 1$, a nonnegative Taylor expansion gives

$$\frac{F_{\ell,q}}{a_{\ell,q}} \preceq z^{2q} H_{\ell-2q} \preceq z^2 H_{\ell-2} \preceq H_\ell. \quad (4.4)$$

Combining (4.2)–(4.4), every nonconstant monomial $\mathbf{y}^\alpha$ in the rook expansion satisfies, for $1 \leq d \leq n-2$,

$$\frac{[\mathbf{y}^\alpha] h_{n-d}}{[\mathbf{y}^\alpha] h_n} \leq 2^d \frac{n-2}{n-2+d} \binom{n-2+d}{2d} \leq \frac{n-d}{n} \frac{n^d}{d!} \binom{n}{d}. \quad (4.5)$$

The last inequality follows after taking the ratio of the two sides and using the central-binomial estimate $\binom{2d}{d}/2^d \geq 2^d/(2d+1)$, with $d = 1, 2, 3, 4, 5$ checked directly and monotonicity thereafter.

Equation (4.5), inserted into (2.2), gives for every $\alpha \neq 0$

$$[\mathbf{y}^\alpha] \text{per}(uA + (1-u)J_n) \leq u[\mathbf{y}^\alpha] \text{per}(A). \quad (4.6)$$

It remains to justify the constant monomial. We record the required common-parameter theorem rather than treating it as a black box. If P, Q are permutation matrices and $A_s = (1-s)P + sQ$,

then A_s satisfies (1.3). After multiplying by permutation matrices, take $P = I$, and write the nontrivial cycles of Q as lengths $a_1, \dots, a_r$. For

$$c_m(v) = 2^{1-m} \sum_q \binom{m}{2q} v^q$$

we first prove the one-cycle base used by the induction. Suppose that Q has one nontrivial cycle of length $\ell \geq 2$, all other points are fixed, and $A_s = (1-s)I + sQ$. Set $y = (1-2s)^2$, expand the rook coefficients as $h_k(y) = \sum_q h_{k,q} y^q$, and put

$$\beta_{k,q} = \frac{(n-k)! h_{k,q}}{n^{n-k} \binom{n}{k}}, \quad p = h_{n,0} = 2^{1-\ell}.$$

The case $n = 2$ is separate: then $\ell = 2$, $c_2(y) = (1+y)/2$, and

$$(B_0, B_1, B_2) = \left(\frac{1}{2}, \frac{1}{2}, c_2(y) \right), \quad \Phi_2 = \frac{1}{2} + \frac{y}{2} u^2 \leq \frac{1}{2} + \frac{y}{2} u \quad (0 \leq u \leq 1).$$

Thus both the chord inequality and the auxiliary coefficient bound hold in dimension two. In the rest of the one-cycle argument assume $n \geq 3$.

The Bernstein transform in (2.6) shows that it is enough to prove

$$\beta_{k,q} \leq \frac{k}{n} h_{n,q} \quad (q \geq 1), \quad \beta_{k,0} \leq c_n + (p - c_n) \frac{k(k-1)}{(n-1)(n-2)}. \quad (4.7)$$

For the first inequality begin with the full cycle $\ell = n$. If $N = n - 2q$, $a_r = h_{2q+r,q}$, and $A_0 = 2n - 2q - 1$, two Chu–Vandermonde sums give

$$\frac{a_{r+1}}{a_r} = \frac{N - r}{r + 1} \frac{A_0 - 2r}{A_0 - r}.$$

Thus, with $d = n - k$ and $T_d = \beta_{n-d,q}/h_{n,q}$,

$$\frac{T_d}{T_{d-1}} = \frac{d(N - d + 1)(n + d - 1)}{n(n - d + 1)(2q + 2d - 1)} \leq \frac{n - d}{n - d + 1}.$$

Indeed, $N - d + 1 \leq n - d$ and $d(n + d - 1) \leq n(2q + 2d - 1)$. Since $T_0 = 1$, induction proves the first inequality in (4.7). Adding one fixed point multiplies the rook polynomial by $1 + z$; if $B_{n,D,q} = \beta_{n-D,q}$, then

$$B_{n+1,D,q} = \frac{D^2 n^{D-1}}{(n+1)^{D+1}} B_{n,D-1,q} + \frac{n+1-D}{n+1} \left(\frac{n}{n+1} \right)^D B_{n,D,q}.$$

Substitution of the already proved bounds leaves $n^2((n+1)^D - n^D) + D(n-D)n^D \geq 0$, so the first inequality persists with any number of fixed points.

For the constant term in the full-cycle case, write $\rho = 2^{n-1} c_n$, $R_d = \beta_{n-d,0}/p$, and

$$Q_d = \rho + (1 - \rho) \frac{(n-d)(n-d-1)}{(n-1)(n-2)}.$$

The even-cycle matching formula gives

$$R_d = \frac{2^d n(d!)^2 (n+d-1)!}{n^d n! (2d)!}, \quad \frac{R_{d+1}}{R_d} = \frac{(d+1)(n+d)}{n(2d+1)}.$$

After clearing positive denominators, the comparison with Q_{d+1}/Q_d has the sign of

$$E_d = (1 - \rho)[-d^3 + (2n - 1)d^2 + 2n] + d[-n^2 + 5n - \rho(2n + 2)].$$

The bounds $\rho \leq 4(n-1)/n^2$ for $n \geq 3$ and $\rho < 3n/(n^2+2)$ for $n \geq 7$ show that E_d first decreases and then increases, with $E_1 \leq 0$ and $E_{n-2} \geq 0$; the cases $3 \leq n \leq 6$ follow by direct substitution. Starting from $R_1 = Q_1 = 1$ before the sign change and from $R_{n-1} = Q_{n-1} = \rho$ after it proves $R_d \leq Q_d$.

The fixed-point recurrence reduces the remaining comparison to an affine function of $c_n/p \in [0, 1]$. At its two endpoints it is respectively

$$(n+1)^{D+1}(n-2) \geq n^D[n(n-1) - D(n-D)]$$

and

$$L + \left(\frac{n}{n+1}\right)^n (1-L) \geq \left(\frac{n}{n+1}\right)^D H,$$

where $L = (n+1-D)(n-D)/(n(n-1))$ and $H = (n(n+1-D) + D^2)/(n(n+1))$. Bernoulli's inequality proves both. This establishes the second inequality in (4.7), including $\ell = 2$ after the direct $n = 3$ base case. Averaging (4.7) against the binomial weights in (2.6) proves the chord inequality for one cycle and arbitrary fixed points.

We now merge cycles. For $a \geq b \geq 2$,

$$c_a(v)c_b(v) = c_{a+b}(v) + \lambda_b(1-v)^b d_{a,b}(v), \quad (4.7a)$$

where

$$(\lambda_b, d_{a,b}) = \begin{cases} (4^{-b}, c_{a-b}), & a > b, \\ (2 \cdot 4^{-b}, 1), & a = b. \end{cases}$$

The first term has one fewer cycle. For the second, if D is the lower-dimensional remaining cycle polynomial and $m = n - 2b$, the rook polynomial satisfies

$$R_{(1-v)^b D}^{(n)}(z, y) = (1-y)^b z^{2b} R_D^{(m)}(z, y). \quad (4.7b)$$

Here, for a cycle polynomial F in dimension N , write

$$R_F^{(N)}(z, y) = \sum_{k=0}^N h_k(y) z^k, \quad B_k^{(N)}(F; y) = \frac{(N-k)! h_k(y)}{N^{N-k} \binom{N}{k}},$$

and

$$\Phi_N(F; u, y) = \sum_{k=0}^N \binom{N}{k} B_k^{(N)}(F; y) u^k (1-u)^{N-k}.$$

If $m = 0$, (4.7b) gives $R_{(1-v)^b}^{(2b)} = (1-y)^b z^{2b}$; hence every coefficient below degree $2b$ vanishes, the top coefficient is $(1-y)^b$, and

$$\Phi_{2b}((1-v)^b; u, y) = u^{2b} (1-y)^b \leq u(1-y)^b.$$

This proves both required properties directly. Assume henceforth that $m \geq 1$. For $k = 2b + r$, (4.7b) also gives the exact lift

$$B_{2b+r}^{(n)}((1-v)^b D; y) = (1-y)^b B_r^{(m)}(D; y) \left(\frac{m}{n}\right)^{m-r} \frac{\binom{m}{r}}{\binom{n}{2b+r}}, \quad (4.7c)$$

while the left side is zero for $k < 2b$. The product of the last two factors is at most one because $\binom{m}{r} = \binom{m}{m-r} \leq \binom{n}{m-r} = \binom{n}{2b+r}$. Consequently the lower-dimensional bound $B_r^{(m)}(D; y) \leq D(y)$ lifts to the corresponding bound in dimension n .

For the chord property, the case $u = 0$ is immediate. For $u > 0$, write $\rho = u + m(1-u)/n$ and $v = u/\rho$. Termwise substitution in the Bernstein sum gives

$$\Phi_n((1-y)^b D; u, y) = (1-y)^b u^{2b} \rho^m \Phi_m(D; v, y). \quad (4.7d)$$

The simultaneously maintained coefficient bound makes $\Phi_m(D; v, y)$ a Bernstein convex combination bounded by $D(y)$ for all $0 \leq v \leq 1$, and $u^{2b} \rho^m \leq u$ for $0 \leq u \leq 1/2$. Equations (4.7a)–(4.7d) therefore merge two cycles while preserving both the chord inequality and the auxiliary coefficient bound. The base cases are the one-cycle result just proved and the identity matrix. Strong induction on r proves the common-parameter theorem.

The coefficient of $\mathbf{y}^0$ in the present independent-parameter problem is exactly that common-parameter case. Since $0 \leq y_i \leq 1$, summing it with (4.6) over all nonconstant monomials completes the proof. $\square$

Theorem 4.2 (shared-center cycle simplex). *Suppose $r \geq 2$ cycles C_i of lengths $\ell_i \geq 2$ share exactly one common center and have pairwise disjoint private supports. Allow $f \geq 0$ additional fixed points, put*

$$n = 1 + f + \sum_i (\ell_i - 1),$$

and assume $a_i \geq 0$ and $\sum_i a_i \leq 1$. If

$$A = I + \sum_{i=1}^r a_i (C_i - I),$$

then A satisfies (1.3) for every $n \geq 3$ and every $0 \leq u \leq 1/2$.

Proof. Put $m_i = \ell_i - 1$, $M = \sum_i m_i$, $c = 1 - \sum_i a_i$, and $d_i = 1 - a_i$. The only supported permutation terms are $I, C_1, \dots, C_r$, whence

$$p = \prod_i d_i^{m_i} \left(c + \sum_i a_i (a_i/d_i)^{m_i} \right). \quad (4.10)$$

Weighted AM–GM, followed by the binary entropy bound, gives

$$p \geq \prod_i [a_i^{a_i} d_i^{d_i}]^{m_i} \geq 2^{-M} \geq 2^{1-n}. \quad (4.11)$$

Boundary cases in which a displayed quotient is undefined follow by continuity.

We first handle $n \geq 32$. For the independent row experiment let p be the probability of no collision and q the probability of exactly one lost occupied column. Splitting overloaded columns coefficientwise gives

$$R_A \preceq pB^n + qF_1 + (1-p-q)F_2, \quad F_2 = (1+2z)^2 B^{n-4}. \quad (4.12)$$

Let D count the double occupancies in private columns. The conflict graph of these events is the line graph of a subdivided star. The weighted tree determinant therefore factors its probability generating function as

$$\mathbb{E}x^D = \prod_j (1 - \lambda_j^2 + \lambda_j^2 x), \quad 0 \leq \lambda_j^2 \leq 1,$$

and

$$\mu = \mathbb{E}D = \sum_i m_i a_i (1 - a_i). \quad (4.13)$$

Consequently, for $\mu \geq 1$,

$$q \leq \Pr(D \leq 1) \leq \mu e^{1-\mu}. \quad (4.14)$$

If $p \geq 2\Phi_I - c_n$, convexity of the unit-matrix radial permanent already closes the chord. Otherwise (4.10), the entropy estimate, and $2^n \Phi_I \leq C_n = 2 + \sqrt{\pi n/2}$ give

$$\mu > \frac{n-1}{4} - \frac{1}{2} \log C_n.$$

For $n \geq 32$, (4.14) implies $q < 1/32$. Exact evaluation of the positive functional Φ gives

$$0 \leq \Phi(F_1) \leq z_n, \quad 0 \leq \Phi(F_2) < \frac{7}{16} z_n, \quad \Phi_I < \frac{1}{1024}.$$

The midpoint margin in (4.12) is therefore greater than

$$z_n \left(\frac{1}{2} - \frac{1}{1024} - \frac{7}{16} - \frac{1}{32} \right) = \frac{31}{1024} z_n.$$

Lemma 2.2 restores the whole half interval.

For $16 \leq n \leq 31$, use the following Bernoulli-splitting lemma. If X is a finite sum of independent Bernoulli variables with $\mu = \mathbb{E}X \geq 2$, then

$$\Pr(X \leq 1) \leq (1 + \mu)e^{-\mu}. \quad (4.14a)$$

Indeed, replacing a Bernoulli parameter t by two independent parameters $t/2$ changes the tail by $(t^2/4)(\Pr(Y=1) - \Pr(Y=0)) \geq 0$. If $\Pr(Y=0) > 0$, then

$$\frac{\Pr(Y=1)}{\Pr(Y=0)} = \sum_r \frac{s_r}{1-s_r} \geq \sum_r s_r = \mathbb{E}Y \geq 1;$$

if $\Pr(Y=0) = 0$, the comparison is immediate. Iterating the split and passing to the Poisson limit proves (4.14a). In the low-permanent region, (4.11) and $2^n \Phi_I \leq 2 + \sqrt{\pi n/2}$ imply $\mu > 11/4$, already at $n = 16$, and hence $q \leq (1 + \mu)e^{-\mu} < 1/4$.

For a polynomial P define

$$\Gamma_j(P) = 2^{-j} \sum_{k=0}^j \frac{\binom{j}{k}}{\binom{n}{k}} \frac{(n-k)!}{n^{n-k}} [z^k] P.$$

Applying this positive transform to (4.12), the derivative of the layer margin with respect to p is $j/(2n) - \Gamma_j(B^n) + \Gamma_j(F_2) > 0$: convexity of the unit-matrix sequence gives $\Gamma_j(B^n) \leq \Phi_I < 1/1024$, whereas $j/(2n) \geq 1/62$. Thus the worst admissible values are $p = z_n$ and $q = 1/4$, and every Bernstein layer reduces to

$$\left(1 - \frac{j}{2n}\right) c_n + \frac{j}{2n} z_n - \Gamma_j \left(z_n B^n + \frac{1}{4} F_1 + (3/4 - z_n) F_2 \right) > 0. \quad (4.15)$$

All 376 rational inequalities (4.15) are verified exactly by the certificate described in the supplement.

For $8 \leq n \leq 15$, keep the endpoint and collision mean coupled:

$$p \geq z_n \exp\{(n-1)/2 - 2\mu\}. \quad (4.16)$$

An extremal argument for a Bernoulli sum gives, for $\mu \geq 3/2$,

$$q \leq \max\{e^{1-\mu}, (1+\mu)e^{-\mu}\}. \quad (4.17)$$

To prove this bound, maximize $\Pr(X \leq 1)$ on the closed fixed-mean cube section and choose a maximizer with the fewest interior coordinates. Varying two such coordinates at fixed sum shows that all remaining interior coordinates are equal; the others are zero or one. One deterministic one gives $e^{1-\mu}$. With no deterministic one, the equal-parameter tail is

$$f_N = (1 - \mu/N)^{N-1}(1 + \mu - \mu/N).$$

For $\mu < N$, the negative power series for $\log(1-x)$ gives

$$\log \frac{f_N}{(1+\mu)e^{-\mu}} \leq \frac{\mu^2}{N} \left[\frac{1}{1+\mu} - \frac{N-1}{2N} - \frac{(N-1)\mu}{3N^2} - \frac{(N-1)\mu^2}{4N^3} \right]. \quad (4.17a)$$

The bracket decreases with μ . For $N \geq 5$, its first two terms already give a nonpositive bound when $\mu \geq 3/2$. For $N = 3, 4$, including the cubic term gives respectively $-2/45$ and $-11/160$ at $\mu = 3/2$; for $N = 2$, including the quartic term gives $-29/640$. The endpoint $\mu = N$ follows by continuity. Thus the tail is at most $(1+\mu)e^{-\mu}$. Two deterministic ones give zero. This proves (4.17).

For each transformed layer put

$$\begin{aligned} A_j &= (1 - j/(2n))c_n - \Gamma_j(F_2), \\ B_j &= j/(2n) - \Gamma_j(B^n) + \Gamma_j(F_2), \\ C_j &= \Gamma_j(F_1) - \Gamma_j(F_2). \end{aligned} \quad (4.18)$$

The margin is at least $A_j + pB_j - qC_j$. Exact scalar algebra gives $B_j > 0$ and $2z_n B_j \geq C_j$. If $\mu \leq 3/2$, (4.16) and $q \leq 1$ leave the gate

$$A_j + z_n B_j E_{12}((n-1)/2 - 3) - C_j > 0,$$

where $E_{12}(x) = \sum_{k=0}^{12} x^k/k!$. If $3/2 \leq \mu \leq (n-1)/4$, the two expressions obtained from (4.17),

$$A_j + z_n B_j e^{(n-1)/2 - 2\mu} - C_j e^{1-\mu},$$

and

$$A_j + z_n B_j e^{(n-1)/2 - 2\mu} - C_j (1+\mu)e^{-\mu},$$

are nonincreasing in μ , precisely because $2z_n B_j \geq C_j$. It is therefore enough to check the endpoint $\mu = (n-1)/4$. Using the rational lower bound $E_{12} \leq e^x$, and $e < 11/4 \leq 1 + (n-1)/4$ for $n \geq 8$, so that at the terminal point the pure-exponential tail is no larger than the Poisson tail, the full continuous domain reduces to the four exact rational gates

$$B_j > 0, \quad 2z_n B_j \geq C_j, \quad A_j + z_n B_j E_{12}((n-1)/2 - 3) - C_j > 0,$$

$$A_j + z_n B_j - C_j \frac{1 + (n-1)/4}{E_{12}((n-1)/4)} > 0. \quad (4.19)$$

All four gates for the 92 layers are checked exactly by the certificate in the supplement; the preceding reduction, rather than the computation alone, covers every continuous weight and every $u \in [0, 1/2]$.

It remains to treat $3 \leq n \leq 7$. For fixed long branches, fixed total weight of length-two branches, and fixed u , the actual chord margin is affine in the product of any two short-branch weights. At a minimum on the short-branch simplex, all positive short-branch weights may therefore be taken equal; otherwise one moves along a fixed-sum line until a weight vanishes without increasing the margin. All-transposition stars are then covered for $3 \leq n \leq 6$ by ten exact continuous certificates. For $n = 7$, the same equal-weight reduction gives the exact minimum

$$p \geq q_6 := \frac{5^5}{6^6} > \frac{1}{32} = 2^{2-7}.$$

Indeed, for k positive equal weights $a \leq 1/k$, $p_k(a) = (1-a)^{k-1}(1-(k+1)a+2ka^2)$ decreases on that interval and has minimum $q_k = (k-1)^{k-1}/k^k$; q_k decreases with k , so $k \leq 6$ gives the displayed bound. Theorem 1.1 now closes the whole chord for $n = 7$. If at least one branch is longer, integer partitions of $\sum_i (\ell_i - 1) \leq 6$ leave exactly 32 support structures. The stick-breaking substitution

$$b_i = x_i \prod_{j < i} (1 - x_j), \quad u = y/2, \quad 0 \leq x_i, y \leq 1,$$

maps a unit box onto each complete weight simplex. Exact Bernstein subdivision proves the margin nonnegative on all 318 leaf boxes. An independent checker reconstructs every coefficient from the original matrix, and the complete propositions and hashes appear in the supplement.

The ranges $3 \leq n \leq 7$, $8 \leq n \leq 15$, $16 \leq n \leq 31$, and $n \geq 32$ are disjoint and exhaustive. They include zero weights, $\sum_i a_i = 1$, and all $f \geq 0$. This proves the theorem. $\square$

5 Cycle–element incidence forests

We now prove Theorem 1.3.

5.1 Compatible cycle selections and an entropy endpoint

The incidence-forest hypothesis implies that every nontrivial directed cycle in the support is one of the original C_i : a simple directed cycle crossing two blocks would have to revisit a cut element. Thus a permanent term is specified by a set of pairwise element-disjoint original cycles.

Let $B_i \in \{0, 1\}$ record whether cycle i is selected. At an element v , the local variables are mutually exclusive, with probabilities a_i for incident cycles and residual probability

$$d_v = 1 - \sum_{i \ni v} a_i.$$

In every component choose one element vertex as root. Direct every incidence edge away from it. Each cycle vertex owns all its child elements; because the incidence graph is a tree, a cycle of length ℓ_i owns exactly $\ell_i - 1$ elements, while the remaining root element accounts for the component itself. The local marginals have a compatible joint law. Propagate away from the

root: if a parent cycle is active, every child cycle at the next element is inactive; if it is inactive, choose one child j with conditional probability $a_j/(1 - a_{\text{parent}})$, or choose none with probability $d_v/(1 - a_{\text{parent}})$. The resulting law q satisfies

$$q(B) = \frac{\prod_v \mu_v(B_{\text{inc}(v)})}{\prod_i \mu_i(B_i)^{\ell_i - 1}}. \quad (5.1)$$

The Gibbs variational inequality $\log \sum_B w_B \geq H(q) + \mathbb{E}_q \log w_B$ then yields

$$p := \text{per}(A) \geq \exp \left(- \sum_i (\ell_i - 1) h(a_i) \right) \geq 2^{-M}, \quad (5.2)$$

where $M = \sum_i (\ell_i - 1)$ and $h(a) = -a \log a - (1 - a) \log(1 - a)$. Since $n = c + f + M$, if $c + f \geq 2$ then $p \geq 2^{2-n}$. Theorem 1.1 proves the first part of Theorem 1.3.

We shall also use the strengthened endpoint–mean relation

$$p \geq \exp \left(-M \log 2 + \frac{M}{2} - 2\mu \right), \quad \mu = \sum_i (\ell_i - 1) a_i (1 - a_i). \quad (5.3)$$

It follows from $\log 2 - h(a) \geq 2(a - 1/2)^2$. Boundary weights follow by continuity.

5.2 A real collision statistic

Use the ownership just defined. Each non-root element v has a unique parent cycle $i(v)$; let $p(v)$ be the predecessor row of v along that directed cycle. For independently chosen rows define

$$D = \sum_v \mathbf{1}_{\{X_{p(v)}=v\}} (N_v - 1). \quad (5.4)$$

If $d = \sum_v (N_v - 1)_+$ is the total collision deficit, then $0 \leq D \leq d$, and $\Pr(d = 1) \leq \Pr(D \leq 1)$. For a child element owned by cycle i , the event $X_{p(v)} = v$ has probability a_i , while the expected number of other rows entering v is $1 - a_i$. Row independence therefore gives

$$\mathbb{E} D = \sum_i (\ell_i - 1) a_i (1 - a_i) = \mu. \quad (5.5)$$

5.3 One-hot decorrelation

We need a tail estimate which does not falsely treat the collision indicators as independent.

Lemma 5.1. *Suppose each row independently selects exactly one column. If f_v is positive, depends only on the indicators entering column v , and is decreasing in every coordinate, then*

$$\mathbb{E} \prod_v f_v \leq \prod_v \mathbb{E} f_v. \quad (5.6)$$

Proof. Condition on all but one row. Write the local values as $a_v \leq b_v$ according as the row selects v or not, and put $\delta_v = 1 - a_v/b_v$. The conditional expectation under the one-hot row equals

$$\left(\prod_v b_v\right) \left(1 - \sum_v p_v \delta_v\right).$$

Replacing the row indicators by independent Bernoulli variables with the same marginals gives

$$\left(\prod_v b_v\right) \prod_v (1 - p_v \delta_v),$$

which is larger because $\prod_v (1 - y_v) \geq 1 - \sum_v y_v$. Replace rows one at a time. At the end the column families are independent, giving (5.6). $\square$

Apply the lemma to $f_v = x^{D_v}$, $0 < x < 1$, where $D_v = \mathbf{1}_{\{X_{p(v)}=v\}}(N_v - 1)$. For each owned column, conditioning on the parent-row event and using $1 - y \leq e^{-y}$ together with $1 - e^{-b} \geq b/2$ for $0 \leq b \leq 1$, gives

$$\mathbb{E}x^{D_v} \leq \exp\left(-\frac{1}{2}a_i(1 - a_i)(1 - x)\right).$$

Multiplying over the $\ell_i - 1$ owned columns of every cycle gives

$$\mathbb{E}x^D \leq \exp\left(-\frac{\mu}{2}(1 - x)\right). \quad (5.7)$$

Writing $\nu = \mu/2$, assume $\nu \geq 1$ and choose $x = 1/\nu$. Then

$$q := \Pr(d = 1) \leq \Pr(D \leq 1) \leq \nu e^{1-\nu}. \quad (5.8)$$

5.4 The large-order chord estimate

Assume $n \geq 64$. Let

$$\Phi_I = \text{per}\left(\frac{I_n + J_n}{2}\right), \quad S_n = 2^n \Phi_I, \quad C_n = 2 + \sqrt{\frac{\pi n}{2}}.$$

If $p \geq 2\Phi_I - c_n$, then

$$\text{per}(uA + (1 - u)J_n) \leq \text{per}(uI_n + (1 - u)J_n)$$

by the coefficientwise unit-matrix envelope. Convexity of the right-hand side on $0 \leq u \leq 1/2$ bounds it by the chord joining c_n and Φ_I , which is below the chord joining c_n and p . Thus this high-permanent range is closed. Otherwise, (5.3) and $S_n \leq C_n$ imply

$$\mu > L_n := \frac{n-1}{4} - \frac{1}{2} \log C_n. \quad (5.9)$$

The function L_n is increasing, and $L_{64} > 13$, so $\nu > 13/2$. Hence

$$q < \frac{13}{2} e^{-11/2} < \frac{1}{32}. \quad (5.10)$$

Use the two-collision envelope

$$R_A \preceq pB^n + qF_1 + (1 - p - q)F_2, \quad F_2 = (1 + 2z)^2 B^{n-4}. \quad (5.11)$$

For $n \geq 32$, direct scalar estimates give

$$0 \leq \Phi(F_2) < \frac{7}{16}z_n, \quad 0 \leq \Phi(F_1) \leq z_n, \quad \Phi_I < \frac{1}{1024}. \quad (5.12)$$

For reference, direct summation yields

$$\frac{\Phi(F_2)}{z_n} = \frac{(4n^2 - n - 6)S_n - 2n(8n - 11)}{2(n-1)(n-2)(n-3)}.$$

Using $S_n \leq 2 + \sqrt{\pi n/2}$ gives the stated $7z_n/16$ bound. The midpoint chord margin therefore satisfies

$$G > z_n \left(\frac{1}{2} - \frac{1}{1024} - \frac{7}{16} - \frac{1}{32} \right) = \frac{31}{1024}z_n > 0. \quad (5.13)$$

The positive Bernstein reduction restores the full interval. This proves Theorem 1.3 for $n \geq 64$. The remaining single-cycle and shared-center cases are Theorems 4.1 and 4.2. $\square$

6 Reproducibility and finite certificates

The proof has three finite exact components:

1. the 495 endpoint layers for $2 \leq n \leq 31$;
2. the 376 Poisson-boundary layers and 92 coupled small-mean layers used in the shared-center theorem;
3. the 32 small support structures, 318 Bernstein leaf boxes, ten low-degree star certificates, and 255 reconstructed exact coefficients.

All use rational arithmetic. Independent checkers reconstruct the relevant rook polynomials and Bernstein leaves from the original matrices; destructive controls are included for the star certificates. The precise proposition certified by each computation, the distinction between discovery and independent checkers, the commands, and the frozen SHA-256 values are recorded in [the finite-certificate supplement](#). The complete internal dependency manifest is recorded in `loops/LIH-WANG-PUBLICATION-PREP-0001/A1-写作前审计包.md`.

These calculations certify finite symbolic leaves. They do not test a finite sample of matrices as a substitute for a general argument.

7 Relation to prior work and limitations

The original Lih–Wang conjecture concerns all doubly stochastic matrices. Theorem 1.1 instead gives a sufficient scalar gate, while allowing the larger row-stochastic domain. These are different kinds of statements. Likewise, star matrices are required to satisfy a convexity inequality against every doubly stochastic comparison matrix over the full unit interval. A necessary permanent lower bound for a star is not, by itself, a sufficient star criterion; Theorem 1.1 should not be read that way.

Goldwasser’s direct-sum theorem concerns radial monotonicity of all-order subpermanents under stronger blockwise hypotheses. At order n , radial monotonicity gives an endpoint comparison

rather than the chord upper bound (1.3). Corollary 1.2 is therefore not a reformulation of that theorem.

Known direct overlaps include the original order-three result, the complete order-four doubly stochastic half interval, a conditional order-six range, and several 2×2 -block star subdomains. We exclude these from any novelty claim. The exact cycle-forest ranges in Theorem 1.3 were not found in the original sources examined, but this statement is bounded by the available literature: the full texts of Lih–Wang (1982), Chang (1983), Goldwasser (1992), and Subramanian–Somasundaram (2016) were not all available to us. Publisher abstracts and later exact restatements were used to bound their main theorems. Accordingly, we do not claim global priority.

The following remain open here:

- the full Lih–Wang conjecture;
- optimality or necessity of 2^{2-n} ;
- connected multi-center cycle forests below order 64;
- incidence structures containing a cycle;
- an unconditional direct-sum transfer theorem.

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